



Uptake and accumulation of per- and polyfluoroalkyl substances in plants

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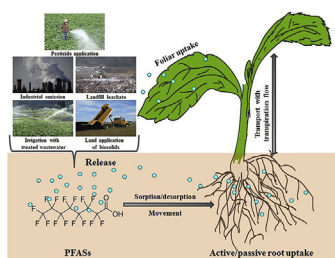
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HIGHLIGHTS

- Up-to-date knowledge on plant uptake and accumulation of PFASs is reviewed.
- Multiple sources of PFASs contribute to their accumulation in plants.
- Root uptake is the major pathway for accumulation of PFASs in agricultural crops.
- Uptake and accumulation of PFASs vary with plants, chemicals and abiotic factors.
- Further mechanistic understanding of plant uptake of PFASs is warranted.

GRAPHICAL ABSTRACT



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ABSTRACT

Per- and polyfluoroalkyl substances (PFASs) are a class of persistent organic contaminants that are ubiquitous in the environment and have been found to be accumulated in agricultural products. Consumption of PFAS-contaminated agricultural products represents a feasible pathway for the trophic transfer of these toxic chemicals along food chains/webs, leading to risks associated with human and animal health. Recently, studies on plant uptake and accumulation of PFASs have rapidly increased; consequently, a review to summarize the current knowledge and highlight future research is needed. Analysis of the publications indicates that a large variety of plant species can take up PFASs from the environment. Vegetables and grains are the most commonly investigated crops, with perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS) as the most studied PFASs. The potential sources of

Abbreviations: AFFFs, Aqueous firefighting foams; BCF, Bioconcentration factor; EtFOSA, N-Ethyl perfluorooctane sulfonamide; EtFOSAA, N-Ethyl perfluorooctane sulfonamidoacetic acid; EtFOSE, N-Ethyl perfluorooctane sulfonamido ethanol; FOSA, Perfluorooctane sulfonamide; FOSAA, Perfluorooctane sulfonamidoacetic acid; FOSA, Perfluoroalkyl sulfonamide; FOSE, Perfluoroalkyl sulfonamide ethanol; FTCA, Fluorotelomer saturated carboxylate; FTUCA, Fluorotelomer unsaturated carboxylate; MeFOSA, N-Methyl perfluorooctane sulfonamide; MeFOSAA, N-Methyl perfluorooctane sulfonamidoacetic acid; MeFOSE, N-Methyl perfluorooctane sulfonamide ethanol; n:2 Cl-PFAES, n:2 Chlorinated polyfluoroalkyl ether sulfonate; n:2 diPAP, n:2 Polyfluoroalkyl phosphate diester; n:2 FTOH, n:2 Fluorotelomer alcohol; n:2 FTSA, n:2 Fluorotelomer sulfonic acid; PFECHS, Perfluoro-4-ethylcyclohexane sulfonate; PFASs, Per- and polyfluoroalkyl substances; PFBA, Perfluorobutanoic acid; PFBS, Perfluorobutanesulfonic acid; PFCA, Perfluoroalkyl carboxylic acids; PFDA, Perfluorodecanoic acid; PFDoDA, Perfluorododecanoic acid; PFDS, Perfluorodecanesulfonic acid; PFHpA, Perfluoroheptanoic acid; PFHpS, Perfluoroheptanesulfonic acid; PFHxA, Perfluorohexanoic acid; PFHxDA, Perfluorohexadecanoic acid; PFHxS, Perfluorohexanesulfonic acid; PFNA, Perfluorononanoic acid; PFNS, Perfluorononanesulfonic acid; PFOA, Perfluorooctanoic acid; PFODA, Perfluorooctadecanoic acid; PFOS, Perfluorooctanesulfonic acid; PFPeA, Perfluoropentanoic acid; PFPeS, Perfluoropentanesulfonic acid; PFPrA, Pentafluoropropionic acid; PFSA, Perfluoroalkyl sulfonic acid; PFTeDA, Perfluorotetradecanoic acid; PFTrDA, Perfluorotridecanoic acid; PFUnDA, Perfluoroundecanoic acid; RCF, Root concentration factor; TF, Translocation factor; TFA, Trifluoroacetic acid; WWTP, Wastewater treatment plant; Σ PFASs, Total concentration of PFASs.

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PFASs for plant uptake include industrial emissions, irrigation with contaminated water, land application of biosolids, leachates from landfill sites, and pesticide application. Root uptake is the predominant pathway for the accumulation of PFASs in agricultural crops, and uptake by plant aboveground portions from the ambient atmosphere could play a minor role in the overall PFAS accumulation. PFAS uptake by plants is influenced by physicochemical properties of compounds (e.g., perfluorocarbon chain length, head group functionality, water solubility, and volatility), plant physiology (e.g., transpiration rate, lipid and protein content), and abiotic factors (e.g., soil organic matters, pH, salinity, and temperature). Based on literature analysis, the current knowledge gaps are identified, and future research prospects are suggested.

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1. Introduction

Per- and polyfluoroalkyl substances (PFASs) are a large group of synthetic organic chemicals containing a hydrophilic head functional group and a hydrophobic $-CF_2$ tail chain. The binding between C–F forms an extremely strong covalent bond (~ 110 kcal/mol), which confers excellent chemical and thermal stability to PFASs (Buck et al., 2011). Since the late 1940s, more than 5000 PFASs have been synthesized and extensively used in a myriad of industrial and household products such as aqueous firefighting foams (AFFFs), surfactants, adhesives, coatings, lubricants, food packages, paints, and pesticides (Prevedouros et al., 2006; Paul et al., 2008; Buck et al., 2011). The widespread applications of these products have led to the ubiquitous distribution of PFASs in the environment and in biota (Benskin et al., 2012; Chen et al., 2018; Dalahmeh et al., 2018; Vuong et al., 2018). Many PFASs, e.g., perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS), have been determined to be persistent in the environment, bioaccumulative, and toxic to animals and humans (Buck et al., 2011; Itoh et al., 2016; ASTDR, 2018). Consequently, PFOS and PFOA, have been listed in the annex to the Stockholm Convention on Persistent Organic Pollutants. The environmental prevalence and potential toxicity of PFASs render them high-priority contaminants of increasing concern.

Agricultural lands are important environmental reservoirs for PFASs. PFASs can enter soils via multiple pathways including land application of biosolids, irrigation with PFAS-contaminated water, leaching from landfilled wastes, release from industrial manufacturing and pesticide application (Taniyasu et al., 2013; Blaine et al., 2014a; Gallen et al., 2016; Liu et al., 2016; Tian et al., 2018). In some highly contaminated sites, the concentration of PFASs in soils reached the mg/kg levels (Jin et al., 2015; Braunig et al., 2019). Due to their relatively high water solubility, large proportions of PFASs can remain dissolved in soil water, rendering them highly mobile and bioavailable to plant roots for uptake (Zhao et al., 2012; Zhu and Kannan, 2019). Plants grown in contaminated areas are frequently detected with measurable concentration of PFASs, confirming the capability of plants to absorb these contaminants from soils (Blaine et al., 2013; Wen et al., 2014; Jin et al., 2018; Yu et al., 2018; Kim et al., 2019). Considering that agricultural crops constitute a major portion of livestock and human diets, consumption of the PFAS-laden food represents an important route responsible for the accumulation of PFASs in animals and humans (van Asselt et al., 2011; Blaine et al., 2014a; Perez et al., 2014; Brambilla et al., 2015). Exposure to PFASs could lead to various health issues including hepatic damage, cardiovascular disease, dysfunction of the immune system, endocrine disorders, reproductive and developmental problems, and cancer (ASTDR, 2018). Therefore, in order to better assess the potential risks associated with human consumption of PFAS-laden crops, it is imperative to clearly elucidate the transfer behaviors of PFASs from

environmental media to plants and their distributions in edible plant structures.

Although there have been an increasing number of studies conducted on PFAS uptake by plants in the fields or in controlled environments (Blaine et al., 2013; Stahl et al., 2013; Felizeter et al., 2014; Wen et al., 2014), review papers focusing on this topic are scarce (Ghisi et al., 2018). Therefore, there is a need to collate the current literatures on the bioaccumulation of PFASs in plants in order to guide future research direction. The objective of this review is to summarize the current research progress on plant uptake and accumulation of PFASs and assess the potential sources, uptake pathways, and factors that influence the accumulation and distribution of PFASs in plants. Based on the reviewed literatures, we highlight the current knowledge gaps and recommend several perspectives for future research.

2. Literature search and analysis

2.1. Data sources

A bibliographic search was conducted using ISI Web of Science (<http://apps.webofknowledge.com>) to retrieve papers published during the interval from January 2009 to December 2019. The keywords used in the search included “perfluoroalkyl substances” or “perfluorinated substances” in combination with “plant”. In addition, a complementary search was also performed by examining the reference lists of the retrieved literatures. The candidate publications were carefully checked to determine whether they contained relevant content. A total of fifty publications were eventually selected and summarized in Tables 1–3.

2.2. Analysis of retrieved literatures

Among the fifty selected publications regarding plant uptake of PFASs, sixteen studies were conducted as field experiments (Table 1), and thirty-one studies under controlled experimental settings, including seventeen pot studies (Table 2) and fourteen hydroponic studies (Table 3). Two studies were conducted via integrating field and greenhouse pot experiments (Blaine et al., 2013; Lee et al., 2014), and one study employed both pot and hydroponic experimental approaches (Zhao et al., 2018b).

Most field studies focused on the accumulation of PFASs in plants grown near a specific point source such as industrial manufacturing sites, fire training fields, landfills, or wastewater treatment plants (WWTPs) (Table 1). The results from these studies indicate the impacts of human activities to PFAS emissions (Gobelius et al., 2017; Liu et al., 2017a; Dalahmeh et al., 2018; Tian et al., 2018). Five studies indicated PFASs can be released from biosolids to soils and migrate into plants (Yoo et al., 2011; Blaine et al., 2013; Lee et al., 2014; Wen et al., 2014; Gottschall et al., 2017). Other studies reported the detection of PFASs in the

samples randomly collected from the fields, and in the vegetables and/or fruits acquired from local markets (Herzke et al., 2013; Sznajder-Katarzynska et al., 2018; Liu et al., 2019a). Out of the publications utilizing the pot experiment, there are eleven studies in which PFASs were spiked directly into soils as the source for plant uptake (Table 2). Soil amendment with biosolids represents another approach for plant exposure to PFASs in pot studies. Other exposure scenarios include irrigation with PFAS-contaminated water and utilization of soils collected from PFAS-contaminated sites (Blaine et al., 2014a; Braunig et al., 2019). Compared with soil-based pot experiments, hydroponic study provides a clear insight into the mechanism of chemical uptake by plants because of the minimal interferences from sorption/desorption, transport, and transformation of target chemicals in soils (Li et al., 2018). The primary exposure method in hydroponic experiment is to cultivate plants in nutrient solution spiked with a predesigned amount of PFASs (Table 3).

Perfluoroalkyl carboxylic acids (PFCAs), particularly those with chain length of 3–11 $-\text{CF}_2$ units, are the most studied PFASs in the retrieved publications. For instance, PFOA, one of the predominant PFCAs in the environment, appears in forty-one studies out of the fifty selected publications. Long-chain PFCAs such as perfluorotridecanoic acid (PFTrDA), perfluorotetradecanoic acid (PFTeDA), perfluorohexadecanoic acid (PFHxDA), and perfluorooctadecanoic acid (PFODA), were also widely detected in plant tissues collected from the contaminated fields (Gobelius et al., 2017; Dalahmeh et al., 2018); however, they were rarely used in pot or hydroponic plant uptake experiments. Perfluoroalkyl sulfonic acids (PFSAAs) with 4–10 carbon chain length were frequently selected as target compounds in field and laboratory studies, and PFOS is the most commonly studied compound. PFAS precursors represent many majors classes of PFASs and there is a current lack of knowledge on their structure/composition and how to evaluate their environmental occurrence, transformation and accumulation in plants (Prevedouros et al., 2006; Zhang et al., 2016; Tian et al., 2018; Zhao et al., 2018c). PFAS precursors such as fluorotelomer alcohols (FTOHs), perfluoroalkyl sulfonamides (FOSAs), perfluoroalkyl sulfonamido ethanols (FOSEs) and fluorotelomer sulfonic acids (FTSAs), have been found in plants grown at contaminated sites (Dalahmeh et al., 2018; Tian et al., 2018); however, few studies have been conducted to evaluate the uptake of these precursor compounds by plants in laboratory settings. Many PFCA precursors including polyfluoroalkyl phosphate diesters (diPAPs), fluorotelomer saturated carboxylates (FTCAs), and fluorotelomer unsaturated carboxylates (FTUCAs) are also rarely investigated in either field or laboratory studies (Lee et al., 2014; Bizkarguenaga et al., 2016b; Tian et al., 2018).

A large variety of plant species have exhibited the capability to take up PFASs via their roots from soil and water, and/or via their aboveground portions from the atmosphere. The majority of studies have focused on uptake by vegetables, followed by grains and fruits, with wheat, maize, lettuce and tomato being the most investigated crop species in both field and laboratory studies (Tables 1–3). The diversity of plant species used in pot or hydroponic studies is much less than those documented in field investigations. A total of twenty-five types of vegetables and grains were reported to accumulate PFASs in the field studies, whereas only thirteen species were used in pot experiments and six in hydroponic settings. Grasses, which constitute an important portion of livestock forage, were also reported to take up PFASs, but the resultant risks to animal and human health were rarely evaluated (Stahl et al., 2009; Garcia-Valcarcel et al., 2014; Wen et al., 2016). In addition to root uptake from soils, tree leaves and bark were found to sorb vapor-phase and particle-bound PFASs, confirming the potential of plant aboveground portions to take up airborne PFASs (Jin

et al., 2018; Tian et al., 2018). Although PFASs were frequently detected in fruits such as apple, banana, orange, strawberry, and lemon collected from local markets, few attempts were made to track the contamination sources in these products (Sznajder-Katarzynska et al., 2018).

3. Plant uptake

3.1. Sources of PFASs in plants

There are several sources contributing to the accumulation of PFASs in plants. Emissions from industrial manufacturing can contaminate the surrounding soil, water, air, and consequently plants. It was reported that the total concentration of PFASs in soil, precipitation, surface water, and groundwater close to a mega fluorochemical industrial park reached up to 641 $\mu\text{g}/\text{kg}$, 4.86 $\mu\text{g}/\text{L}$, 1860 $\mu\text{g}/\text{L}$, and 273 $\mu\text{g}/\text{L}$, respectively (Liu et al., 2016; Liu et al., 2017a). In China, approximately 80–90% of PFOS/PFOA contamination can be tracked back to industrial manufacturing (Liu et al., 2017b). These PFASs in the environment can enter plants in the surrounding areas of the manufacturing sites. For instance, agricultural crops grown around fluorochemical industrial parks were found to contain high levels of PFASs; the total concentration was up to 87 $\mu\text{g}/\text{kg}$ wet weight (ww) in vegetables, 480 $\mu\text{g}/\text{kg}$ dry weight (dw) in wheat grains, and 59 $\mu\text{g}/\text{kg}$ dw in maize grains, which were attributed primarily to the nearby industrial discharges (Liu et al., 2016; Liu et al., 2017a; Bao et al., 2019). In addition, tree bark and leaves collected near a fluorochemical manufacturing park were also found to be contaminated with PFASs, suggesting that the airborne PFASs released from industries could be sorbed by the aboveground portions of plants (Jin et al., 2018).

Field irrigation with PFAS-contaminated water represents an important pathway responsible for PFAS accumulation in plants. The massive production and long-term use of PFASs has led to their ubiquitous presence in freshwater bodies such as lakes, rivers, and groundwater (Guo et al., 2015; Lu et al., 2015; Pan et al., 2018). Approximately 70% of freshwater resources are used in agricultural production (Pimentel et al., 2004). In some highly contaminated sites, e.g., the areas surrounding fluorochemical manufacturing facilities, the total concentration of PFASs was measured at mg/L level in nearby surface waters (Chen et al., 2018; Bao et al., 2019). Comparable levels of PFASs were also found in surface water and groundwater close to the sites where AFFFs had been frequently used such as at airports, firefighter training sites, and military bases (Backe et al., 2013; Gobelius et al., 2017; Dauchy et al., 2019). In China, the annual release of PFASs to nearby surface waters from the applications of AFFFs was estimated to be 3.5 tons of PFOS and 0.35 tons of PFOA (Liu et al., 2017b). In Tyndall Air Force Base, USA, PFOA was found in groundwater with the concentration up to 6570 $\mu\text{g}/\text{L}$ (Moody and Field, 1999). Surface runoff and infiltration from these AFFF-applied sites can lead to considerable PFAS contamination in soils and plants (Gobelius et al., 2017; Braunig et al., 2019). The amphiphilic nature of PFASs enables large portions of these chemicals to remain dissolved in the effluents of WWTPs (Kim et al., 2016). It was reported that the average concentration of the total PFASs in the effluents from three WWTPs in Guangzhou, China was between 0.016 and 0.234 $\mu\text{g}/\text{L}$, and the daily mass loads of PFASs to rivers ranged between 1.55 and 25.1 g (Pan et al., 2016). In Australia, the annual emission of PFOA and PFOS from the effluents of WWTPs is up to 65 and 26 kg, respectively (Gallen et al., 2018). Increasing use of treated wastewater in field irrigation consequently introduces more PFASs to agricultural production systems (Blaine et al., 2014a).

Land application of biosolids is a common practice to improve soil quality, and is another major pathway to introduce PFASs into

Table 1
Summary of field studies on plant accumulation of PFASs.

Plant species	Contamination sources	Compounds	Concentration in plants	Reference
Tomato, cucumber, eggplant, pepper, & Chinese cabbage	Fluorochemical industrial park	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFBS, PFHxS, PFOS	∑PFASs: 1.7–87 µg/kg ww.	Bao et al. (2019)
Camphor, cypress, & magnolia	Fluorochemical industrial park	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFTTrDA, PFTeDA, PFBS, PFHxS, PFOS	∑PFASs: Tree leaves: 46–1115 µg/kg dw; Tree bark: 37–774 µg/kg dw.	Jin et al. (2018)
Wheat & maize	Fluorochemical industrial park	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFBS, PFHxS, PFOS	∑PFASs: Wheat grain: 1.13–480 µg/kg dw; Maize grain: 0.7–58.8 µg/kg dw.	Liu et al. (2017a)
Radish, carrot, Chinese cabbage, Chinese chives, lettuce, Welsh onion, celery, cauliflower, pepper, pumpkin, wheat, maize, & soybean	Fluorochemical industrial park	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFBS, PFHxS, PFOS	∑PFASs: 1.36–8085 µg/kg dw.	Liu et al. (2019b)
Sedge, Canada wildrye, & sumac	Fluorochemical industrial park	PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA	Average ∑PFASs: Grass roots: 61 µg/kg dw; Grass leaves: 68 µg/kg dw; Tree leaves: 430 µg/kg dw.	Zhu and Kannan (2019)
Rice	Industrial complexes and wastewater treatment plant	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFTTrDA, PFTeDA, PFHxDA, PFODA, PFBS, PFHxS, PFOS, PFDS	∑PFASs: 0–1.85 µg/kg dw.	Kim et al. (2019)
Yam, sugarcane, & maize	Wastewater treatment plant	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFTTrDA, PFTeDA, PFHxDA, PFOcA, PFBS, PFHxS, PFOS, PFDS, PFOSA, MeFOSA, EtFOSA, MeFOSE, EtFOSE, MeFOSAA, EtFOSAA, 6:2FTSA	∑PFASs: Yam root: 0.36 ± 0.17 µg/kg dw; Sugarcane stem: 0.35 ± 0.064 µg/kg dw; Maize cob: 0.2 ± 0.064 µg/kg dw.	Dalahmeh et al. (2018)
Silver birch, Norway spruce, bird cherry, mountain ash, ground elder, long beechfern, & wild strawberry	Fire Training Facility	PFHxS, PFOS, PFDS, PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFTTrDA, PFTeDA, PFHxDA, PFODA, PFBS, PFHxS, PFOSA, MeFOSA, EtFOSA, MeFOSE, EtFOSE, FOSAA, MeFOSAA, EtFOSAA, 6:2 FTSA	∑PFASs: Silver birch: 6.6–337 µg/kg dw; Norway spruce: 1.4–222 µg/kg dw; Bird cherry: 2.0–58 µg/kg dw; Mountain ash: 0.3–8.2 µg/kg dw; Ground elder: 2.9–83 µg/kg dw; Long beechfern: 3.1–35 µg/kg dw; Wild strawberry: 5.8–8 µg/kg dw.	Gobelius et al. (2017)
Lettuce & tomato	Biosolids	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFBS, PFHxS, PFHpS, PFOS, PFDS	Lettuce: up to 27.5 µg/kg dw (PFBA); Tomato: up to 17.0 µg/kg dw (PFBA).	Blaine et al. (2013)
Pumpkin	Biosolids	diPAPs and PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA	∑diPAPs: 33.98 µg/kg ww; ∑PFCAs: 50.75 µg/kg ww.	Lee et al. (2014)
Maize, soybean, & wheat	Biosolids	PFHxS, PFOS, PFDS, PFOSA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA	No PFAS was detected.	Gottschall et al. (2017)

Table 1 (continued)

Plant species	Contamination sources	Compounds	Concentration in plants	Reference
Wheat	Biosolids	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFTTrDA, PFTeDA, PFBS, PFHxS, PFHpS, PFOS, PFDS	∑PFASs: Root: 140 –472 µg/kg dw; Straw: 36.2 –178 µg/kg dw; Husk: 6.15 –37.8 µg/kg dw; Grain: 7.32 –35.6 µg/kg dw.	Wen et al. (2014)
Tall fescue, barley, bermuda grass, & Kentucky bluegrass	Biosolids	PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFTTrDA, PFTeDA, PFBS, PFHxS, PFOS	Average ∑PFASs: Up to 709.1 µg/kg dw.	Yoo et al. (2011)
Chinese pine, oriental plane, & abele	Landfill sites	TFA, PFPrA, PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFBS, PFHxS, PFOS, 6:2 diPAP, 8:2 diPAP, 6:2 FTOH, 8:2 FTOH, 10:2FTOH, MeFOSE, EtFOSE, MeFOSA, EtFOSA	Average ∑PFASs: >6200 µg/kg dw in tree leaves.	Tian et al. (2018)
Canadian poplar, Chinese scholartree, weeping willow, / & Chinese red pine		PFHxA, PFHpA, PFOA, PFNA, PFDA, PFDoDA, PFTeDA, PFBS, PFPeS, PFHxS, PFHpS, PFOS, PFNS, PFDS, MeFOSAA, EtFOSAA, 4:2 FTSA, 6:2 FTSA, 8:2 FTSA, 6:2 Cl-PFAES, 8:2 Cl-PFAES, PFECHS	Tree bark: ∑PFASs: 10 µg/kg dw; ∑PFASs: 0.65 µg/kg dw; Cl-PFAESs: <0.15 µg/kg dw; PFECHS: <0.11 µg/kg dw; FTSAs: <1.7 µg/kg dw; FOSAs: <0.51 µg/kg dw.	Liu et al. (2019a)
Carrot, onion, tomato, zucchini, cucumber, eggplant, pepper, cauliflower, cabbage, lettuce, spinach, chicory, asparagus, celery, fennel, potato, pea, & bean	/	PFHxA, PFHpA, PFOA, PFNA, PFDA, PFDoDA, PFUnDA, PFODA, PFOS, PFTTrDA, PFTeDA, PFHxS	Average ∑PFASs: 10.5–38.1 µg/kg ww.	Herzke et al. (2013)
Eucalyptus	Pesticide application	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFTTrDA, PFTeDA, PFBS, PFHxS, PFOS, PFDS, FOSA, FOSAA, EtFOSA, EtFOSAA	∑PFASs: 0.979 µg/kg dw in leaves	Nascimento et al. (2018)
Banana, apple, lemon, orange, cherry, strawberry, potato, beetroot, carrot, white cabbage, & tomato	/	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFBS, PFHxS, PFOS	∑PFASs: Up to 49.6 µg/kg ww.	Sznajder-Katarzynska et al. (2018)

agricultural fields (Blaine et al., 2014a; Wen et al., 2014). As described above, PFASs can survive most municipal or industrial wastewater treatments and remain in significant amounts in the sewage sludge i.e., biosolids (Alder and van der Voet, 2015). PFAS concentration in biosolids can reach µg/kg to low mg/kg levels (Schultz et al., 2006; Loganathan et al., 2007). Widespread and continuous land application of biosolids can lead to the accumulation of PFASs in soils. For example, farmlands in Decatur, Alabama, USA have received applications of biosolids for more than a decade, and the soils contained the total PFAS concentration of up to several micrograms per gram of dry soils (Washington et al., 2010). It was estimated that the total annual loading of PFASs from the land application of biosolids to agricultural soils in the USA is 1.4–2.1 tons (Venkatesan and Halden, 2013). PFASs in soils can be taken up by agricultural crops posing high health risks to animals and humans via food consumption (Lindstrom et al., 2011; Blaine et al., 2014a; Lee et al., 2014; Bizkarguenaga et al., 2016a).

Leachates from landfill sites have been identified as a point source contributing to the dissemination of PFASs in the environment (Paul et al., 2008). In northern Spain, the maximum total concentration of 11 PFASs in leachates from four solid waste landfill sites was up to 3.2 µg/L (Fuentes et al., 2017). In China, the total amount of PFAS leakages from landfills to soils and the surrounding groundwater was estimated to be 3.1 tons per year (Yan et al., 2015). PFASs in landfill leachates may enter soil and water, and further migrate to plants (Liu et al., 2017b; Hepburn et al., 2019). Landfills

can also release vapor-phase and particle-bound PFASs, as well as their precursor compounds, to the ambient air (Ahrens et al., 2011). For example, the total concentration of PFASs was up to 9.5 ng/m³ in the air, and 4100 µg/kg in the dry deposition around two landfill sites in Tianjin, China (Tian et al., 2018). Atmospheric emissions of PFASs from solid waste landfills in Ontario, Canada were approximately 1 kg per year (Ahrens et al., 2011). PFASs present in the ambient atmosphere of landfills were diverse, usually with 8:2 fluorotelomer alcohol (8:2 FTOH) and perfluorobutanoic acid (PFBA) as the dominant PFASs (Ahrens et al., 2011; Weinberg et al., 2011; Tian et al., 2018). Plant leaves and bark can sorb these airborne PFASs, leading to the accumulation of PFASs in above-ground portions of plants (Jin et al., 2018; Tian et al., 2018).

N-Ethyl perfluorooctane sulfonamide (EtFOSA) is the active ingredient of sulfluramid, a widely used pesticide for controlling leaf-cutting ants and cockroaches (Lofstedt Gilljam et al., 2016; Nascimento et al., 2018). As a precursor compound, EtFOSA can be decomposed into several derivatives including perfluorooctane sulfonamide (FOSA) and PFOS (Plumlee et al., 2009; Lofstedt Gilljam et al., 2016; Zhao et al., 2018c). The extensive application of sulfluramid in agricultural production has resulted in considerable dissemination of PFASs into the environment (Lofstedt Gilljam et al., 2016; Liu et al., 2017b). From 2004 to 2015, Brazil released 147.3 tons of FOSA and 19.5 tons of PFOS into the environment due to the use of sulfluramid (Lofstedt Gilljam et al., 2016). In China, emissions of PFOA and PFOS from the application of sulfluramid

Table 2
Summary of pot experiment studies on plant uptake of PFASs.

Plant species	Compounds	Soil treatment	Initial concentration	Cultivation time	Observations	Reference
Alfalfa, Italian ryegrass, lettuce, maize, mung bean, radish, & soybean	EtFOSAA	Soil + spiked biosolids	258 ± 21 pmol/g dw.	>60 days	EtFOSAA was only accumulated in plant roots with BCFs in the range of 0.52–1.37 pmol/g soil; EtFOSAA metabolites including EtFOSA, FOSAA, FOSA, and PFOS were detected in plant roots and aerial parts. Plant concentration of EtFOSAA correlated closely with the protein and lipid contents.	Wen et al. (2018)
Carrot, potato, & cucumber	PFOA & PFOS	Soil + spiked sewage sludge	PFOA: 276–805 µg/kg dw; PFOS: 10–556 µg/kg dw.	63–96 days	Vegetative compartments concentrated more PFASs than peelings and peeled edible parts for all the plants; PFOA exhibited a higher transfer potential than PFOS.	Lechner and Knapp (2011)
Carrot & lettuce	8:2 diPAP	Soil + spiked compost (95:5, w/w)	500 µg/kg dw.	30–90 days	8:2 diPAP (4.6–7.8 µg/kg) and its metabolites including PFNA (1.0–1.3 µg/kg), PFOA (12–157 µg/kg), PFHpA (3–44 µg/kg), PFHxA (2.9–144 µg/kg), PFPeA (3.7–115 µg/kg), PFBA (3.9–130 µg/kg) were detected in root peels, root core, and leaves of carrot; 8:2 diPAP and PFOA were detected in lettuce heart; The total BCFs of 8:2 diPAP and its metabolites were <0.004 and 0.64–32.60 respectively in carrot, and 0.04–0.18 and 0.28–1.57 respectively in lettuce heart.	Bizkarguenaga et al. (2016b)
Lettuce & tomato	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFBS, PFHxS, PFHpS, PFOS, PFDS	Soil + biosolids (90:10, w/w)	∑PFASs: 335–434 µg/kg dw.	56–112 days	PFBA and PFPeA were the most accumulated PFASs with the concentration of 266 and 236 µg/kg respectively in lettuce, and 56 and 211 µg/kg respectively in tomato; The BCFs of PFASs were up to 56.8 (PFBA) in lettuce and 17.1 (PFPeA) in tomato.	Blaine et al. (2013)
Radish, celery, tomato, & sugar snap pea	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFBS, PFHxS, PFOS	Soil + biosolids	∑PFASs: 329.13–427.36 µg/kg dw.	67–224 days	PFOA (67 µg/kg) was the highest congener in celery shoot, while PFBA was the most accumulated in radish roots (232 µg/kg) and pea fruit (150 µg/kg); PFBA exhibited the highest BCFs in shoots of all crops; Fruit crops accumulated fewer long-chain PFASs than shoot or root crops.	Blaine et al. (2014a)
Alfalfa	6:2 diPAP	Soil + biosolids	167 µg/kg dw.	5.5 months	∑PFASs were up to 58,000 µg/kg ww in plant; Short-chain PFCAs dominated the accumulated PFASs in plant; The BCFs of PFASs in plant inversely correlated their carbon chain lengths.	Lee et al. (2014)
Twenty lettuce cultivars	PFOA	Soil spiked with PFOA solution	200–1000 µg/kg.	50 days	PFOA exhibited different effects on root and shoot biomass of different lettuce cultivars; The accumulation and translocation of PFOA varied significantly with different cultivars; PFOA was mainly distributed in the plant cell walls, followed by organelles.	Xiang et al. (2018)
Twenty-six lettuce cultivars	PFOS	Soil spiked with PFOS solution	200–1000 µg/kg.	45 days	Significant variation in the biomass of shoots and roots was observed among different lettuce cultivars; Different cultivars exhibited large variation in accumulating and translocating PFOS; Protein contents and protein-mediated transpiration related closely with PFOS accumulation in the lettuce cultivars.	Yu et al. (2018)
Maize	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFBS, PFHxS, PFOS	Soil spiked with PFASs solution	250–1000 µg/kg dw for individual PFAS.	128 days	PFBA and PFPeA were the highest in maize straw and kernel, respectively; Short-chain PFASs predominated in both maize straw and kernel; PFASs were preferentially accumulated in maize straw relative to the kernel.	Krippner et al. (2015)
Wheat	PFBA, PFHxA, PFOA, PFBS, PFHxS, PFOS	Soil spiked with PFASs solution	200–2000 µg/kg dw.	4 weeks	Higher exposure dose led to reduced shoot biomass in wheat; Exposure to PFBA and PFBS reduced <i>Chlorophyll a</i> content in wheat while PFOA and PFOS exhibited promoting effect; RCFs and TFs for short-chain PFASs were much higher than those for long-chain PFASs.	Lan et al. (2018)
Wheat & rapeseed	PFOA & PFOS	Soil spiked with PFASs solution	PFOA: 285–292 µg/kg dw; PFOS: 264–294 µg/kg dw.	70 days	Plant biomass was increased by 15.2–31.0%; The contents of chlorophyll and malondialdehyde and activities of superoxide dismutase and peroxidase were decreased in wheat, but were increased in rapeseed;	Zhao et al. (2017)

Table 2 (continued)

Plant species	Compounds	Soil treatment	Initial concentration	Cultivation time	Observations	Reference
Wheat	PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFBS, PFHxS, PFOS	Soil spiked with PFASs solution	200–1000 µg/kg.	30 days	PFAS concentration in plant roots and shoots was in the range of 332–1411 and 39.6–821 µg/kg, respectively. The accumulated concentration of PFASs in wheat roots and shoots increased with exposure dose; Log BCFs of PFASs in wheat decreased linearly with increasing carbon chain lengths; PFHxA exhibited the highest transfer potential in plant.	Zhao et al. (2014)
Wheat	PFOA	Soil spiked with PFOA solution	200–800,000 µg/kg dw.	28 days	Wheat germination rate was decreased as exposure dose increased; Low exposure dose stimulated the seedling growth and root length of wheat, while high exposure exhibited inhibiting effects; Plant proline content and peroxidase activity increased as PFOA exposure concentration increased, but the catalase activity decreased.	Zhou et al. (2016)
Wheat	FOSA	Soil spiked with FOSA solution	2.128 nmol/g dw.	30 days	FOSA and its metabolites including PFOS, PFHxS and PFBS were detected in wheat roots and shoots; BCFs of FOSA in wheat roots were much higher than PFOS.	Zhao et al. (2018b)
Spring wheat, oat, potato, maize, & perennial ryegrass	PFOA & PFOS	Soil spiked with PFASs solution	250–50,000 µg/kg dw.	Up to several months	Higher exposure levels led to higher plant accumulation of PFASs; Vegetative compartments accumulated more PFASs than storage organs; PFOA was more preferentially accumulated in plants except the potato peels.	Stahl et al. (2009)
Wheat, rye, rapeseed, & barley	PFOA & PFOS	Soil spiked with PFASs solution	25,000 µg/kg dw.	/	PFAS concentration in plant straws was much higher than those in grains; Uptake of PFASs generally exhibited a decreasing temporal trend.	Stahl et al. (2013)
Lettuce & strawberry	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFBS, PFHxS, PFOS	Irrigation with spiked reclaimed water	0.2–40 µg/L for individual PFAS.	/	Plant uptake of PFASs increased linearly with the exposure dose of compounds but decreased with their carbon chain lengths; PFCA were more preferentially absorbed by plants than PFASs; BCFs for lettuce decreased 0.4–0.6 log unit per additional –CF ₂ group.	Blaine et al. (2014b)
Alfalfa, Italian ryegrass, lettuce, maize, mung bean, radish & soybean	PFOA & PFOS	Soils collected from biosolids-amended field	PFOA: 416.8 µg/kg dw; PFOS: 154.4 µg/kg dw.	45 days	Root concentration of PFOA and PFOS was 703.4–4310.3 and 212.4–723.6 µg/kg dw, respectively, and their shoot concentration was 85.8–3500.9 and 25.5–105.5 µg/kg dw, respectively; RCFs of PFOS and PFOA were 1.37–4.68 and 1.69–10.3, respectively; TFs of PFOA (0.09–1.78) were much higher than PFOS (0.06–0.179); The accumulation of PFOS and PFOA in roots correlated positively with root protein contents but negatively with root lipid contents.	Wen et al. (2016)
Wheat grass	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFBS, PFHxS, PFOS	Soils collected from airport sites	∑PFASs: 2400–14,000 µg/kg dw.	10 weeks	PFHxS and PFOS were the most accumulated compounds with their plant concentration up to 2800 and 1100 µg/kg ww, respectively; BCFs of PFASs in wheat grass ranged from 0.06 (PFNA)–70 (PFBA); BCFs in grass shoots decreased 0.4–0.7 log units per additional –CF ₂ for PFCA and 0.4–1.9 for PFASs;	Braunig et al. (2019)
Red chicory	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFBS, PFHxS, PFOS	Combined use of contaminated irrigation water and spiked soil	0–80,000 µg/L in irrigation water and 0–200 µg/kg dw in spiked soil.	87 days	PFBA, PFPeA, PFHxA and PFBS were the most accumulated PFASs; PFBA had the highest BCFs in plant root (221), leaf (109.4) and head (38); PFAS concentration and BCFs decreased with their increasing carbon chain lengths in all plant compartments.	Gredelj et al. (2019)

were estimated to be 2.6 and 1.4 tons, respectively, per year (Zhang et al., 2012). Plants can take up EtFOSA and its metabolites through roots and/or foliage, leading to the accumulation of PFASs in plants (Nascimento et al., 2018; Tian et al., 2018; Zhao et al., 2018c).

3.2. Plant uptake pathways

Root uptake from soil and water is considered the primary pathway for PFASs to enter plants (Stahl et al., 2009; Lee et al., 2014; Wen et al., 2014). PFASs are typically comprised of a hydrophobic tail –CF₂ chain and a hydrophilic head functional group, conferring relatively high water solubility (Table S1). PFASs generally

Table 3
Summary of hydroponic studies on plant uptake of PFASs.

Plant species	Compounds	Initial concentration	Exposure time	Observations	Reference
Lettuce	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFTTrDA, PFTeDA, PFBS, PFHxS, PFOS	0.01–10 µg/L	40 days	The uptake isotherms for long-chain PFASs in plant roots were not linear; Long-chain PFASs demonstrated higher RCFs; TFs for PFASs generally decreased with their carbon chain lengths; Long-chain PFASs exhibited better transferability from plant root to foliage.	Felizeter et al. (2012)
Tomato, cabbage, & zucchini	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFTTrDA, PFTeDA, PFBS, PFHxS, PFOS	0.01–10 µg/L	/	PFASs were present in different plant compartments; RCFs for PFASs and PFASs generally increased with carbon chain lengths; Short-chain PFASs (<C10) were more transferable to aerial parts.	Felizeter et al. (2014)
Pumpkin, soybean, & wheat	EtFOSA	1.155 mmol/L	12 h	EtFOSA and its metabolites including FOSAA, FOSA, PFOS, PFHxS, and PFBS were present in plant roots and shoots; Root uptake followed the order: EtFOSA > FOSA PFOS ~ FOSAA > PFBS ~ PFHxS; RCFs of EtFOSA correlated positively with the root lipid contents.	Zhao et al. (2018c)
Pumpkin	6:2 FTSA	1.100 mmol/L	12 days	6:2 FTSA and its metabolites including PFHpA, PFHxA, PFPeA, PFBA, PFPrA, and TFA were accumulated in plant roots and shoots; RCFs of 6:2 FTSA were 2.6–24.2 times higher than those of its metabolites; TFs of 6:2 FTSA and its metabolites were negatively correlated with their logK _{ow} .	Zhao et al. (2019)
<i>Arabidopsis thaliana</i>	PFOA	0–50,000 µg/L	21 days	Exposure to PFOA led to decreased biomass in plant roots and shoots; PFOA concentration in plant shoots was 8.5–12.5 times of those in roots; PFOA exposure significantly increased H ₂ O ₂ and malondialdehyde contents in shoots.	Yang et al. (2015)
Soybean	8:2 FTOH	0.215 mmol/L	12–144 h	The distribution of 8:2 FTOH and its metabolites followed the order of roots > stem > leaves; Glutathione-conjugated metabolites were present in soybean tissues; Activities of some 8:2 FTOH degrading enzymes were increased.	Zhang et al. (2016)
Great brome	PFBA, PFBS, PFHxS, PFOA, PFOS, PFDA	500–1000 µg/L	1–20 days	The concentration of PFASs in plant increased with exposure time; Plant uptake of PFASs decreased with carbon chain lengths; Higher exposure dose led to higher BCFs in plants.	Garcia-Valcarcel et al. (2014)
Wheat	PFBA, PFHpA, PFOA, PFDoDA	1000 µg/L	0–7 days	The concentration of PFASs in wheat roots and shoots increased with exposure time; RCFs increased but TFs decreased with carbon chain length of PFASs; The highest uptake of PFASs by wheat was observed at pH = 7.	Zhao et al. (2018a)
Wheat	PFOS	1000 µg/L	0–7 days	Accumulation of PFOS in the wheat roots and shoots increased with increasing salinity (0.03–7.25 psu), temperature (20–30 °C) and exposure concentration (100–100,000 µg/L); The maximum uptake of PFOS by wheat was observed at pH = 6.	Zhao et al. (2013)
Wheat	PFBA, PFHpA, PFOA, PFDoDA	1000 µg/L	5 days	Plant uptake of PFASs increased by 2.8–4.2 times as the solution salinity increased from 0 to 0.4%; The uptake of PFASs increased by 1.5–2.3 times as the culture temperature increased from 20 to 30 °C.	Zhao et al. (2016a)
Wheat	PFOS	100–200,000 µg/L	7 days	High PFOS exposure (>100,000 µg/L) significantly inhibited the synthesis of plant chlorophyll and soluble protein contents and the biomass of roots and leaves; Plant peroxidase and superoxide dismutase activities were elevated at low exposure dose (<10,000 µg/L) but were inhibited at high exposure levels (>200,000 µg/L).	Qu et al. (2010)
Wheat	FOSA	1.900 mmol/L	12 days	FOSA and its metabolites including PFOS, PFHxS, and PFBS were accumulated in wheat roots and shoots; Plant concentration of PFASs followed order of root > shoot and FOSA > PFOS > PFHxA > PFBS.	Zhao et al. (2018b)
Wheat	TFA, PFPrA, PFBA, PFHxA, PFOA, PFOS	0–25,000 µg/L	2–80 h	Plant root and shoot exhibited different concentration trends with carbon chain lengths of PFASs; Root uptake of PFASs was a carrier-mediated process; Different PFASs followed different uptake pathways.	Zhang et al. (2019)
Maize	PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFBS, PFHxS, PFOS	100 µg/L	5 days	Plant uptake of PFDA decreased significantly with increasing solution pH (5–7); PFBA (2460 µg/kg dw) and PFOS (3630 µg/kg dw) were respectively the most accumulated PFCA and PFSA in maize roots; TFs for both PFASs and PFASs decreased with carbon chain lengths.	Krippner et al. (2014)
Maize	PFOS & PFOA	100–200,000 µg/L	2–100 h	High exposure level (>10,000 µg/L) significantly inhibited the root and shoot biomass; The root uptake of PFOA and PFOS fitted well with Michaelis-Menten model; Plant uptake of PFOA and PFOS might follow different pathways.	Wen et al. (2013)

demonstrated a relatively low sorption potential by soils, especially for the short-chain homologues (Buck et al., 2011; Zhao et al., 2012). PFASs in soils can migrate towards plant roots, driven primarily by the water potential gradient created by plant transpiration and PFAS concentration gradient (Lechner and Knapp, 2011). The uptake mechanism of PFASs into plant roots remains largely unexplored, though some studies have initiated mechanistic research. Wen et al. (2013) observed that when maize roots were exposed to 0.1 mM of sodium azide (NaN_3) and 0.6 mM of sodium vanadate (Na_3VO_4) as metabolic inhibitors, the uptake of PFOA decreased by 83% and 43%, respectively. NaN_3 and Na_3VO_4 treatment could reduce the production of adenosine triphosphate in root cells via inhibiting oxidative phosphorylation and phosphatase activity (Thompson et al., 2000; Krapfenbauer et al., 2003), which caused the decreased uptake of PFOA in plant. Therefore, the authors concluded that PFOA movement into maize roots could be an energy-dependent active process (Wen et al., 2013). However, they found PFOS uptake was insensitive to these two metabolic inhibitors. In the presence of aquaporin inhibitors (e.g., silver nitrate and glycerol) and an anion channel blocker (e.g., 5-nitro 2-(3-phenylpropylamine)benzoic acid), Wen et al. (2013) observed the decreased uptake of PFOS by 25–31% and 30–33%, respectively, and suggested that the uptake of PFOS by maize roots could be a passive process. More recently, Zhang et al. (2019) investigated wheat uptake of trifluoroacetic acid (TFA), pentafluoropropionic acid (PFPrA), PFBA, perfluorohexanoic acid (PFHxA), PFOA, and PFOS. The authors found that the accumulation of PFASs in wheat was reduced by 26–95% in the presence of Na_3VO_4 , and that uptake inhibition rates for all tested PFASs exhibited a dose-response relationship with Na_3VO_4 concentration between 0.06 and 0.6 mM. Zhang et al. (2019) proposed that wheat uptake of PFASs including PFOS was an energy-dependent active process. These studies have provided the initial investigation on the uptake mechanism of PFASs by plants. However, metabolic inhibitors could cause changes in stomatal opening, hydraulic conductivity, and photosynthetic rate (Zelitch, 1965; Brauer and Stitt, 1990; Zhang and Tyerman, 1991), which might influence PFAS uptake. The factors responsible for the differences in PFOS uptake between maize and wheat remain unclear. However, both studies demonstrated that PFAS uptake by plants was a concentration-dependent process, which could be well described by the Michaelis-Menten model, implying that the penetration of PFASs into plant roots was probably mediated by carriers in cell membranes such as aquaporins and anion channels (Wen et al., 2013; Zhang et al., 2019). After penetration through a series of root cellular structures including the epidermis, cortex, and endodermis via apoplastic and symplastic pathways, PFASs can enter the root vascular cylinder and subsequently move upward to plant shoots (Blaine et al., 2013, 2014a). Although some progress has been made, large knowledge gaps still exist regarding the mechanisms involved in the uptake and transport of PFASs in plants.

Vapor-phase and particle-bound PFASs and their precursors can also be sorbed by the aboveground portions of plants such as leaves and bark (Jin et al., 2018; Tian et al., 2018; Liu et al., 2019a). Although most PFASs have a low vapor pressure, they can be bound to atmospheric particulates, and deposit on plant aboveground surfaces (Shan et al., 2014). Fluorochemical manufacturers, WWTPs, and landfills can disseminate a substantial amount of PFASs to the surrounding atmosphere with the concentration reaching ng/m^3 levels in the air and mg/kg of atmospheric particulates (Ahrens et al., 2011; Weinberg et al., 2011; Jin et al., 2018). Short-chain PFAS homologues such as TFA and PFBA, are typically detected at higher concentrations in both the vapor and particulate phase (Scott et al., 2006; Wu et al., 2014; Tian et al., 2018). In a field study conducted at Tianjin, China, the maximum concentration of

23 PFASs in tree leaves collected near landfill sites was reported to be $> 6200 \mu\text{g}/\text{kg dw}$, with TFA as the dominant congener with concentration range of 560–3000 $\mu\text{g}/\text{kg dw}$ (Tian et al., 2018). The accumulation of PFASs in tree leaves was partially attributed to foliar uptake of airborne PFASs and/or metabolism of their precursors, because the dominant species of PFASs in tree leaves were also predominated in the ambient atmosphere. Additionally, the total concentration of long-chain PFASs (C9–C12) in tree leaves was found to be 1.2 to 50 times greater than those of short-chain homologues (C5–C7), suggesting the preference of foliar uptake for long-chain PFASs (Tian et al., 2018). PFASs were also detected in tree bark collected near a fluorochemical manufacturing park, with the total concentration ranging from 37 to 774 $\mu\text{g}/\text{kg dw}$ (Jin et al., 2018). PFHxA was the predominant PFAS congener (17–559 $\mu\text{g}/\text{kg dw}$) in the bark samples, followed by PFOA (4–269 $\mu\text{g}/\text{kg dw}$), PFBA (0–22 $\mu\text{g}/\text{kg dw}$), and PFTeDA (0–64 $\mu\text{g}/\text{kg dw}$). Since tree bark is composed primarily of inactive plant tissues with minimal water flow, the accumulated PFASs in tree bark were speculated to originate mainly from the atmosphere (Jin et al., 2018). In most cases, the contribution from aerial uptake to the overall accumulation of PFASs in plants could be small to minimal, as evidenced by the fact that no PFASs were detected in the aboveground portions of wheat grown in uncontaminated soils located 8 km away from the fields amended with PFAS-contaminated biosolids (Wen et al., 2014). The minor contribution of foliar uptake to plant accumulation of PFASs could be due to the short contact time between vapor-phase and particulate-bound PFASs and plant aboveground surfaces, the resistance of waxy cuticles in foliage, and the relatively low concentration of airborne PFASs (Buck et al., 2011; Collins et al., 2011; Jin et al., 2018).

3.3. Distribution of PFASs in plants

After they pass through the Casparian strip in plant roots, PFASs enter root xylem and move upwards to different plant structures, with the extent of translocation related closely with the transpiration stream. The plant structures which receive larger amount of water to the sites could accumulate more PFASs, as long as the sufficient domains are available for sorption or incorporation of PFASs. Stahl et al. (2009) studied the uptake of PFOA and PFOS by several cultivated plants including spring wheat, oat, potato, maize, and perennial ryegrass, and found that the concentrations of PFOA and PFOS were much higher in the straw of spring wheat, oat, and maize compared to the grain of the same plant. After exposure to 1000 $\mu\text{g}/\text{kg}$ of mixed PFASs in soil for 128 days, the total concentration of PFASs in maize straw (52 $\text{mg}/\text{kg dw}$) was approximately 60 times higher than that in maize kernels (0.86 $\text{mg}/\text{kg dw}$) (Krippner et al., 2015). In a pot study with soils amended with sewage sludge, Lechner and Knapp (2011) found the higher accumulation of PFOA and PFOS in carrot foliage, cucumber leaves and stalks, and potato leaves and stalks, and less amount in the peeled edible parts. As for the distribution of PFASs in trees around Stockholm Arlanda Airport, Sweden, Gobelius et al. (2017) reported that the total concentration of PFASs in birch (*Betula pendula*) followed the order of leaves (12–97 $\mu\text{g}/\text{kg ww}$) > twigs (5.3–40 $\mu\text{g}/\text{kg ww}$) > trunk/core (0.37–31 $\mu\text{g}/\text{kg ww}$) > roots (2.6–6.2 $\mu\text{g}/\text{kg ww}$). A similar trend was also observed for spruce (*Picea abies*) in the order of leaves > twigs > roots > trunk/core for the total PFAS concentration. These studies suggest the existence of preferential accumulation of PFASs in vegetative structures of plants, which could be partly due to enrichment from the greater transpiration stream. Although some experimental data have been collected, the factors and driving force controlling PFAS distribution in plant structures still remain largely unknown.

The distribution of PFASs within plants can be evaluated with

translocation factor (TF) which is calculated as the ratio of PFAS concentration in shoots to that in roots. For a given PFAS, the reported TF values varied in 2–3 orders of magnitude (Fig. 1). Perfluorocarbon chain lengths and head functional groups of PFASs affect the distribution of PFASs within plants (Felizeter et al., 2012; Wen et al., 2016; Lan et al., 2018; Xiang et al., 2018). In a pot study with wheat, the TF values of PFBA, PFBS, PFHxA, and PFHxS were within the range of 1.0–12, much higher than those of PFOA (< 0.5) and PFOS (< 0.1) (Lan et al., 2018). For hydroponically grown lettuce, PFBA and perfluoropentanoic acid (PFPeA) exhibited greater transferability with their TF values > 1 when the exposure dose was 1–10 $\mu\text{g/L}$, while PFCAs with 5–13 units of $-\text{CF}_2$ (C_6 – C_{14}) chain and PFSA with 4–8 units of $-\text{CF}_2$ chain were less transferable in plants with their TF values < 1 (Felizeter et al., 2012). Uptake of PFASs with large TF values leads to the high accumulation in leaves, grains, or fruits. On the other hand, for PFASs with low TF values, the higher accumulation is expected to be in the edible portions of tuber and root crops, such as radish, carrot, and potato.

The accumulation of PFASs in the edible structures of plants may pose health risks to animals and humans via food consumption. For crops grown in soils amended with the biosolids from industrial manufactures, the total concentrations of PFASs in celery shoots, pea fruits, and radish roots were up to 814, 236, and 279 $\mu\text{g/kg dw}$, respectively (Blaine et al., 2014a). Fruits and vegetables collected from local markets in several European countries were frequently found with PFAS contamination (Herzke et al., 2013; Sznajder-Katarzynska et al., 2018). In perennial grasses grown near a fluoropolymer manufacturing facility, the total concentrations of six PFCAs in grass leaves were in the range of 9–540 $\mu\text{g/kg dw}$ (Zhu and Kannan, 2019). As grasses are primarily used as livestock forage, PFASs accumulated in grasses may migrate to animals and eventually to humans via trophic food chains. At a Swedish dairy cattle farm, the daily intake of PFASs by cows from consumption of silage was estimated to be 0.027 $\mu\text{g/kg}$, and the concentration of PFASs in cow tissues and milk was measured up to 0.228 $\mu\text{g/kg}$ and 0.018 $\mu\text{g/L}$, respectively (Vestergren et al., 2013). Consumption of PFAS-contaminated edible crops and transfer through food chains represent important exposure pathways for humans to these chemicals (Domingo and Nadal, 2017).

3.4. Influencing factors for PFAS accumulation in plants

3.4.1. Physicochemical properties of PFASs

The perfluorocarbon chain lengths of PFASs can significantly influence the uptake and transport of these chemicals in plants. Short-chain PFASs typically have relatively higher water solubility and smaller molecular size (Table S1), both of which facilitate their penetration through plant root layers (e.g. the epidermis, cortex, endodermis, and pericycle) to the vascular cylinder. Therefore, short-chain PFASs generally demonstrate a higher accumulation potential in plants compared to long-chain homologues (Fig. 2). Lan et al. (2018) reported that wheat growing in soils containing 200 $\mu\text{g/kg}$ of PFBA accumulated 3711 $\mu\text{g/kg dw}$ in roots and 44,088 $\mu\text{g/kg dw}$ in shoots, which were 20 and 44 times greater than the accumulation of PFOA in roots and shoots, respectively. For lettuce grown in the soils amended with industrial biosolids, short-chain PFASs such as PFBA, PFPeA, and PFBS were found to accumulate in plants with concentration $> 200 \mu\text{g/kg dw}$, whereas their concentration in soils was relatively low, 4.68 $\mu\text{g/kg}$ for PFBA, 11.6 $\mu\text{g/kg}$ for PFPeA, and 48.6 $\mu\text{g/kg}$ for PFBS (Blaine et al., 2013). Although relatively higher concentration was observed for long-chain PFASs in soils, 93.5 $\mu\text{g/kg}$ for perfluorodecanoic acid (PFDA) and 49.7 $\mu\text{g/kg}$ for PFOS, their concentration in the lettuce was measured to be $< 100 \mu\text{g/kg dw}$. For several crops tested, the bio-concentration factors (BCFs) of PFASs ranged from < 1 to several

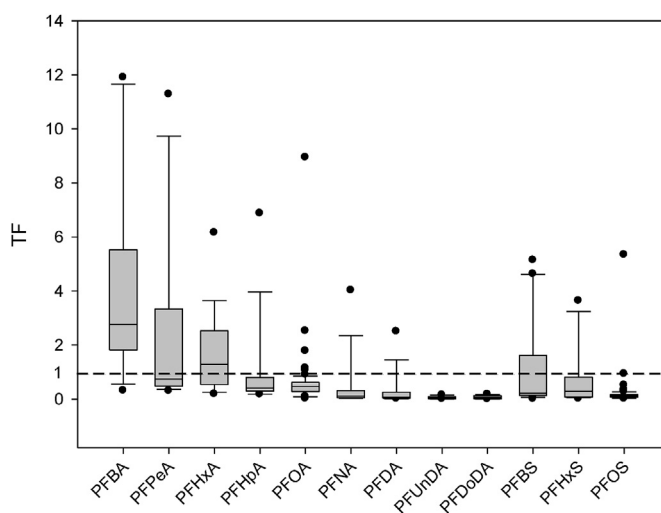


Fig. 1. Translocation factors (TF) of selected PFASs in plants. The TF value was calculated as the ratio of the concentration of a PFAS in plant shoots to that in plant roots. In the box-and-whisker plot, the solid horizontal lines of each box from top to bottom represent the maximum, 75th, median, 25th, and minimum of TF value distribution. The black dots outside the box are outliers. Data are selected from the following references (Felizeter et al., 2012; Blaine et al., 2014a; Zhao et al., 2014, 2017, 2018c; Wen et al., 2016; Lan et al., 2018; Xiang et al., 2018; Yu et al., 2018; Gredelj et al., 2019; Zhang et al., 2019).

thousand depending on PFAS properties, plant species and structures, as well as soil compositions (Blaine et al., 2014a; Lee et al., 2014; Wen et al., 2014; Ghisi et al., 2018). A general trend was found that BCFs decrease by 0.2–0.7 log units with one incremental $-\text{CF}_2$ unit in the alkyl chain (Fig. 2). For long-chain PFASs, sorption by soils limits their movement to plant root rhizosphere (Milinovic et al., 2015; Zhao et al., 2016b). One incremental $-\text{CF}_2$ unit can lead to the increase of sorption coefficients in soils by approximately 0.5–0.6 log units (Higgins and Luthy, 2006). The increased sorption of PFASs, particularly for long-chain homologues, in soils could diminish their bioavailability to plant for uptake.

Hydroponic experiments eliminate the inference from sorption by soils, and the obtained results are better to elucidate the relationship between bioaccumulation in plants and PFAS structures. Krippner et al. (2014) found that PFCA accumulation in the roots of hydroponically grown maize decreased with increasing carbon chain length from C4 (PFBA) to C7 (PFHpA), and then increased with increasing carbon chain length from C7 to C10 (PFDA). A similar U-shape trend was also observed in hydroponically grown wheat (Felizeter et al., 2012). However, for several vegetables such as tomato, cabbage and zucchini, root concentration factor (RCF) generally increased with increasing carbon chain length for both PFCAs (C4–C14) and PFSA (C4–C8) (Felizeter et al., 2014). This could be attributed largely to the higher sorption of longer-chain PFASs on root surfaces (Felizeter et al., 2012, 2014; Krippner et al., 2014).

Compared with long-chain homologues, short-chain PFASs exhibit a higher transferability from plant roots to the aboveground portions. Garcia-Valcarcel et al. (2014) reported that for *Bromus diandrus* exposed to a dosage of 500 $\mu\text{g/L}$ of various PFASs, the concentration of PFBA in the aboveground portions was 3027 $\mu\text{g/kg}$ after exposure for 20 days, which was 2.6 times higher than the measured concentration of PFDA. In hydroponically grown lettuce, grass, and wheat, the TF values of PFASs were found to decrease with increasing $-\text{CF}_2$ chain length (Felizeter et al., 2012; Krippner et al., 2014; Zhao et al., 2018a). The transport of chemicals from plant roots to the aboveground portions occurs in the xylem, and is

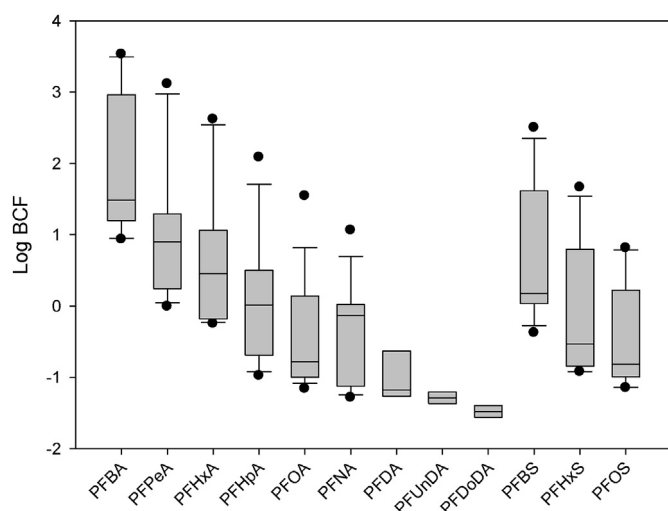


Fig. 2. Logarithmic bioconcentration factors (logBCF) of selected PFASs in plants grown in soils. The BCF value was calculated as the ratio of PFAS concentration in plant to that in soil. In the box-and-whisker plot, the solid horizontal lines of each box from top to bottom represent the maximum, 75th, median, 25th, and minimum of logBCF value distribution. The black dots outside the box are outliers. Data are selected from the following references (Blaine et al., 2013, 2014b; Zhao et al., 2014; Bizkarguenaga et al., 2016b; Braunig et al., 2019).

thought to be driven primarily by the transpiration stream (Collins et al., 2006; Yu et al., 2018). Short-chain PFASs are generally more mobile and less sorptive to plant tissues, conferring higher transferability with transpiration stream when compared to long-chain homologues (Felizeter et al., 2012). This could lead to higher accumulation of short-chain PFASs in the aboveground portions of plants (Felizeter et al., 2012; Krippner et al., 2014). Long-chain PFASs are less efficient in penetrating root surfaces and readily sorbed or blocked by the Casparian strip, and thus preferentially accumulate in plant roots (Felizeter et al., 2012, 2014; Krippner et al., 2015).

The head functional groups of PFASs also influence their accumulation in plants. In the case of PFASs with equal carbon chain lengths, PFCAs demonstrated a higher uptake potential than PFSA. For instance, the accumulation of PFOA in spring wheat, oat, maize, perennial ryegrass, and peeled potato from soils was up to hundreds of times higher than that of PFOS (Stahl et al., 2009). Sorption of PFOS by soil was about 1.7 times more than PFOA (Higgins and Luthy, 2006), which is not sufficient to cause such difference in plant uptake. In a hydroponic study with lettuce, exposed to the same concentration of a mixture of PFASs (1 µg/L), PFBA concentration was 4.83 µg/kg in the roots and 11.7 µg/kg in the leaves, which was 2.7 and 22 times higher than that of PFBS in the corresponding lettuce structures (Felizeter et al., 2012). It has been documented that BCFs of perfluorocarboxylates could be one order of magnitude higher than those of perfluorosulfonate analogs (Yoo et al., 2011; Blaine et al., 2013; Liu et al., 2017a; Dalahmeh et al., 2018; Scher et al., 2018). However, the specific mechanism responsible for the higher accumulation of PFCAs in plants relative to PFSA with equal carbon chain lengths is still unclear.

3.4.2. Plant physiological characteristics

Transpiration plays an important role in controlling the uptake and transport of xenobiotic chemicals in plants (Collins et al., 2011). Difference in transpiration rate could lead to significant variations in PFAS accumulation among plant species, cultivars, plant structures, and even at different growth stages. In a field study conducted near a fluorochemical industrial facility, the total

concentration of 12 PFASs in wheat grains was >11-fold higher than those in maize grains, with short-chain PFCAs (C4 to C8) as the predominant congeners in both crops (Liu et al., 2017a). Compared to maize with transpiration coefficient of 250–300 g water/g dw, wheat possesses a greater transpiration coefficient 450–600 g water/g dw, which is considered as the major contributor to the higher accumulation of PFASs in wheat. In a pot study with 26 lettuce cultivars grown in PFOS-spiked soils, Yu et al. (2018) observed a significant positive correlation between BCF of PFOS in plant shoots and transpiration rate, confirming the important role of transpiration in transporting PFASs in lettuce cultivars. Compared with storage structures (e.g., grain), vegetative structures (e.g., leaf and stalk) tend to accumulate higher levels of PFASs, which is probably due to enrichment from the transpiration stream. In a field study, PFAS accumulation in wheat straw was approximately 5 times more than that of wheat grain (Wen et al., 2014). The BCFs of PFOA and PFOS in the leaves and stalks of potato, carrot, and cucumber ranged between 0.12 and 0.99, significantly higher than those in peelings and peeled edible portions (BCFs < 0.05) (Lechner and Knapp, 2011). Muller et al. (2016) found a positive correlation between the accumulation mass of many PFASs and plant biomass. The increasing biomass with plant growing could provide more retention domains for the accumulation of PFASs.

The bioaccumulation and distribution of PFASs in plants is also dependent on plant composition, especially protein content. Wen et al. (2016) selected maize, soybean, radish, mung bean, lettuce, alfalfa, and Italian ryegrass as test plants with root protein content ranging from 3.35 to 10.4% and lipid content from 2.33 to 5.40%, and found that RCFs of PFOA and PFOS were positively related to root protein contents, but negatively related to root lipid contents. The TFs of PFOA and PFOS also exhibited a positive relationship with the ratio of shoot to root protein content. Similar relationships between RCFs or TFs of PFOS and plant protein contents have also been observed in different lettuce cultivars (Yu et al., 2018). As a major component of carriers or channels in cell membranes, proteins could mediate the transport of PFASs into root cells (Wen et al., 2013; Zhang et al., 2019). Additionally, protein kinases can regulate the formation and development of stomata in plant leaves, which indirectly influences plant transpiration rate (Lai et al., 2005; Yu et al., 2018). The strong association between proteins and PFAS could also enhance the accumulation of PFASs in plants.

3.4.3. Surrounding environments

Sorption of PFASs by soils could reduce their bioavailability to plant roots for uptake. Soil organic matter is considered the major domains influencing the sorption of PFASs by soils (Higgins and Luthy, 2006). A positive relationship has also been observed between sorption coefficient of PFASs in soils and soil organic carbon content (Milinovic et al., 2015). Increasing soil organic matter content can enhance the partitioning of PFASs in soils, decreasing the concentration of PFASs dissolved in soil water for plant uptake. Blaine et al. (2014a) found that the concentration of PFASs in lettuce grown in a soil with 0.4% organic carbon content was 6.5 times greater than that in the lettuce grown in a soil with 6.0% organic carbon content. Long-chain PFASs are more readily sorbed by soils, and demonstrate a lower bioavailability for plant uptake (Zhao et al., 2012, 2016b). Short-chain PFASs have a higher water solubility, exhibit less sorption by soils, and are more mobile in soils. This makes them more bioavailable for plant uptake (Zhao et al., 2012; Milinovic et al., 2015). PFASs containing equal carbon chains but different hydrophilic head functional groups demonstrate varying sorption by soils (Higgins and Luthy, 2006). The difference in sorption of PFASs by soils could be partially responsible for their varying bioaccumulation potentials in plants. Aging can increase the sequestration of PFASs in soils, and further reduce

the bioavailability of PFASs to plant roots for uptake (Blaine et al., 2013; Zhao et al., 2016b).

Other factors such as salinity, pH, and temperature could also influence the accumulation of PFASs in plants. Increasing salinity reduces plant transpiration, which might consequently diminish plant uptake of water, nutrients and xenobiotic contaminants (Stofberg et al., 2015). However, in the two reported hydroponic experiments, increasing solution salinity was found to enhance the accumulation of PFASs in wheat, which might be due to the possible changes in the physicochemical properties of PFASs and/or physiological characteristics of plants (Zhao et al., 2013, 2016a). Krippner et al. (2014) measured PFDA uptake by maize at pH 5, 6 and 7, and found that the highest uptake occurred at pH 5.0. For short-chain PFASs such as PFBA and PFHxA, the uptake by maize increased with increasing pH, but the underlying mechanism is still unknown. Temperature can promote plant physiological activities and biochemical processes such as increasing transpiration and metabolism rates, which in turn influence plant uptake of xenobiotic contaminants (Ekvall and Greger, 2003). Zhao et al. (2016a) demonstrated that the accumulation of many PFASs in wheat increased by 1.5–2.3 times when the exposure temperature increased from 20 to 30 °C. However, such temperature effect was not consistent for all the PFASs tested; the uptake of long-chain PFASs was found to be more sensitive to the change in temperature.

4. Summary and research prospects

The long manufacturing history of PFASs and their widespread use in industrial and household applications, coupled with their highly mobile and persistent natures, have caused their ubiquitous dissemination in the environment. This review summarizes the current knowledge regarding the uptake and accumulation of PFASs in plants. The primary sources responsible for the accumulation of PFASs in plants include the emissions from industrial manufacturing, irrigation with contaminated water, land application of biosolids, and leachates from landfills. A number of field and laboratory studies have demonstrated that plants are capable of taking up PFASs from soil and water, as well as from the atmosphere; however, the fundamental mechanism governing the uptake and transport of PFASs in plants still remains largely unknown, particularly the mechanism of PFAS movement through plant cell membranes and the mechanism that controls PFAS distribution in plants. Although adsorption of PFAS-bound particulates on the aboveground portions of plants or gas exchange into foliage can occur, root uptake is still considered the predominant pathway for PFASs to enter plants. PFOA and PFOS, along with other PFCA and PFSA homologues, are the most investigated compounds in the reported studies. Few studies were conducted to elucidate uptake, transport and metabolism of PFAS precursors in plants, though some precursor compounds such as FTOHs, FOSAs, FOSEs and FTSAAs have been detected in plant samples at contaminated sites. The majority of the studies on plant uptake of PFASs are limited to some common grains and vegetables, and the comparisons are infeasible among different plant species with varying physiological characteristics. To facilitate more accurate evaluation of the risks to human health via dietary consumption of agricultural products containing PFASs, the uptake and transport of PFASs in plants needs more thorough investigations with some suggested research priorities worthy of immediate attention:

- (1) More comprehensive field studies are needed to further identify the exposure pathways and quantify the contributions to the accumulation of PFASs in a broader suite of plant species, especially edible vegetables, fruits and grains. The linkages to human exposure and health via food

consumption need to be thoroughly assessed to warrant the safety of agricultural products.

- (2) Considering that transformation of polyfluoroalkyl precursor chemicals constitutes an important source of PFAS contamination in the environment, the uptake and metabolism of PFAS precursors and their relative contribution to the total accumulation of PFASs in plants should be investigated.
- (3) More studies are needed on the genetic-level understanding of PFAS uptake by plants, as well as the phytotoxic effects of the accumulated PFASs on plant health.
- (4) Develop a holistic model to comprehensively evaluate the influencing factors on PFAS uptake and accumulation in plants including soil properties, plant physiological and biochemical processes, and physicochemical properties of PFASs.

Credit author statement

Wenfeng Wang: analyzed the data, conceptualized the knowledge frame, and wrote the paper, Geoff Rhodes: analyzed the data, conceptualized the knowledge frame, and wrote the paper, Jing Ge: reviewed and edited the paper, Xiangyang Yu: reviewed and edited the paper, Hui Li analyzed the data, conceptualized the knowledge frame, and wrote the paper.

Declaration of competing interest

We confirm that there are no known conflicts of interest associated with this work and there has been no significant financial support for this work that could have influenced its outcome. We confirmed that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us. Due care has been taken to ensure the integrity of the work. No part of this paper has been published or submitted elsewhere.

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Appendix A. Supplementary data

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